Comp683: Computational Biology

Lecture 15

March 23, 2026

Today

- Trajectories with diffusion maps
- Trajectories with SLICER

announcement: project presentations begin on Wednesday. Please check the schedule for your time slot. Please submit one proposal document per group in canvas. Attendance is required on both days.

Entering Pseudotime and Trajectory Inference

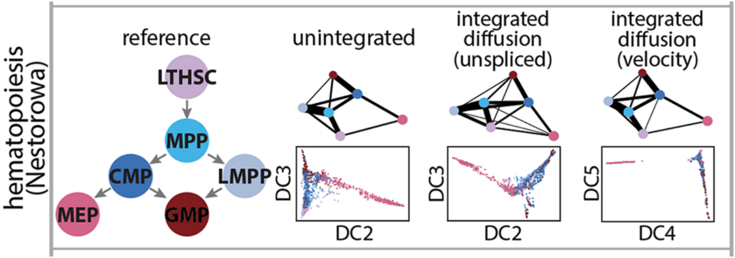

Figure: from Raneket al.Genome Biology. 2022. We would like to establish a chronological ordering of cells that represents their differentiation. This will allow us to uncover gene programs associated with differentiation.

Welcome Back Diffusion Maps

Let's set up some notation as follows :

- Let Ω be the set of all measured cells
- Suppose that we have measured n total cells, each with G features, so a given $\mathbf{x}_i \in \mathbb{R}^G$.

Allow each cell to diffuse around its measured position as,

$$Y_x\left(\mathbf{x}'\right) = \left(\frac{2}{\pi\sigma^2}\right)^{1/4} \exp\left(-\frac{\left\|\mathbf{x}' - \mathbf{x}\right\|^2}{\sigma^2}\right)$$

Here the σ^2 determines the scale over which each cell can randomly diffuse.

Modeling the Probability cell x transitioning to cell y

Define the transition matrix, **P** for all pairs of cells as,

$$P_{\mathrm{xy}} = \frac{1}{Z(\mathbf{x})} \exp\left(-\frac{||\mathbf{x}-\mathbf{y}||^2}{2\sigma^2}\right)$$
$$Z(x) = \sum_{\mathbf{y}\in\Omega} \exp\left(-\frac{||\mathbf{x}-\mathbf{y}||^2}{2\sigma^2}\right)$$

Here $Z(\mathbf{x})$ is a partition function providing an estimate for the number of neighbors of $\mathbf{x}$ in a certain volume defined by σ, or the density of cells at that proximity.

Redefine a Density Normalized Transition Probability Matrix, $\tilde{\mathbf{P}}$

We are interested in the transition probabilities between cells and not the on-cell potentials imposed by local density. Therefore, the diagonal of $\tilde{\mathbf{P}}$ will be set to 0 and $\mathbf{y} = \mathbf{x}$ will be excluded from the sum in the partition function, $\tilde{Z}(\mathbf{x})$. Without some correction, it may look like cells within a very dense region have a small diffusion distance.

$$\tilde{P}_{\mathrm{xy}} = \frac{1}{\tilde{Z}(\mathbf{x})} \frac{\exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right)}{Z(\mathbf{x})Z(\mathbf{y})} \quad , \tilde{P}_{\mathrm{xx}} = 0$$

$$\tilde{Z}(\mathbf{x}) = \sum_{\mathbf{y}\in\Omega/\mathbf{x}} \frac{\exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right)}{Z(\mathbf{x})Z(\mathbf{y})}$$

Using Eigenvectors of $\tilde{\mathbf{P}}$

- $\tilde{\mathbf{P}}$ has n ordered eigenvalues, $\lambda_0 = 1 > \lambda_1 \geq \cdots \geq \lambda_{n-1}$
- Corresponding to these eigenvalues are eigenvectors $\boldsymbol{\Psi}_0 \ldots \boldsymbol{\Psi}_{n-1}$
- As we saw a few weeks ago, powering $\tilde{\mathbf{P}}$ to $\tilde{\mathbf{P}}^t$ represents the probability of transitioning between two cells with a walk of length t.

The diffusion distance between a pair of cells $\mathbf{x}$ and $\mathbf{y}$ can be written in terms of the eigenvectors of $\tilde{\mathbf{P}}$ as,

$$D_t^2(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n-1} \lambda_i^{2t} \left(\psi_i(\boldsymbol{x}) - \psi_i(\boldsymbol{y})\right)^2$$

Unpacking Diffusion Distance

$$D_t^2(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n-1} \lambda_i^{2t} \left(\psi_i(\boldsymbol{x}) - \psi_i(\boldsymbol{y})\right)^2$$

- The eigenvector to the largest eigenvalue, λ_0 is a constant vector, $\boldsymbol{\Psi}_0 = \mathbf{1}$. Therefore, it contributes 0.
- The eigenvalues of $\tilde{\mathbf{P}}$ determine the diffusion coefficients in the direction of the corresponding eigenvector
- After the first l prominent directions, the diffusion coefficients typically drop to a noise level.
- When you find the l such that there is a large difference between l and $l+1$ eigenvalues (an elbow), you can use the sum up to the l-th term as an approximation for diffusion distance. The first l eigenvectors correspond to the diffusion components.

What we have just defined, illustrated

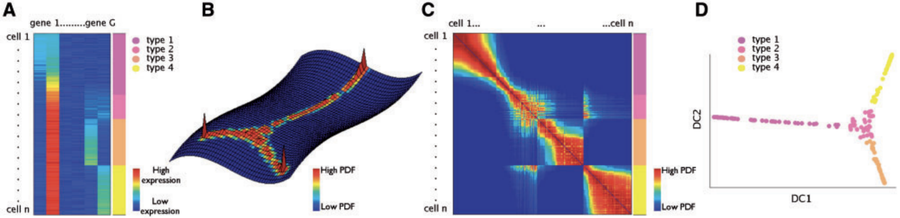

Figure: from Haghverdi *et al.* Bioinformatics. 2015. After projecting cells based on the first two diffusion components, it is still unclear what the time direction is.

Example Applied to A Dataset of Differentiating Cells

The diffusion map is capturing transitions between cell types that are not reflected in PCA or tSNE.

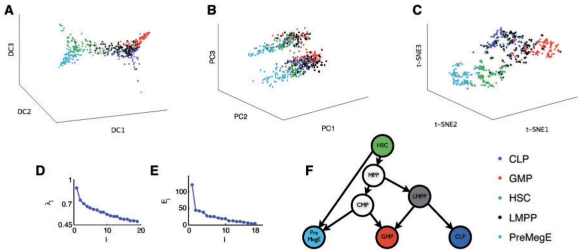

Figure: from Haghverdi *et al.* Bioinformatics. 2015.

Reviewing Diffusion Distance

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This diffusion map approach was the beginning of thousands of people starting to think about cellular differentiation.....

SLICER Intro

SLICER builds on and expands the very early Diffusion based techniques through the following

- Automatically select genes to use for building the trajectory (or in establishing the ordering between cells)
- Use locally linear embedding to capture non-linear relationships between gene expression levels and progression through a process
- Define 'geodesic entropy' and use it to define branches
- Capture unique trajectory patterns such as bubbles.

SLICER Overview

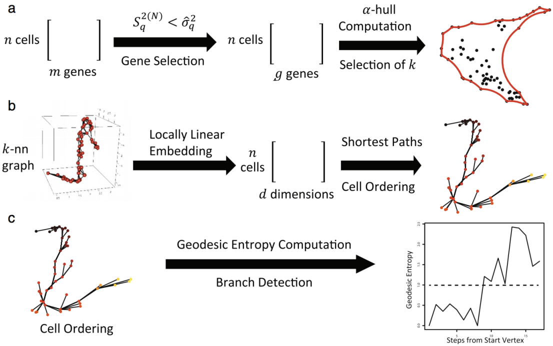


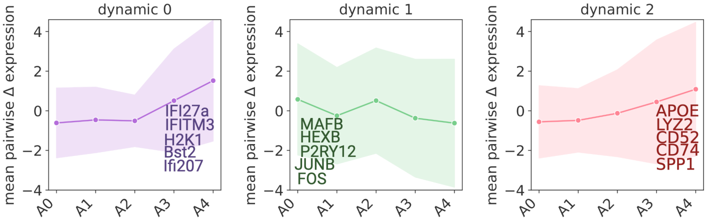
Figure: from Welch *et al.* Genome Biology. 2016

Step 1: Selecting Features to Use (Intuition)

Establishing some intuition about what makes a good ‘trajectory feature‘

- If a feature is involved in progression along a trajectory, expect gradual change in that feature along the trajectory
- A feature not involved should not fluctuate along the trajectory.
- In real life, we have no idea what is happening with this trajectory. Use similarity within neighborhoods to study ‘segments‘ of a trajectory.

Example feature patterns



Age Bins (months)
A0 - [0,6)
A1 - [6,12)
A2 - [12,18)
A3 - [18,24)
A4 - [24,30)

Figure: Example features that have similar expression within certain parts of the trajectory but are dynamic overall.

Neighborhood Variance

Interesting features are those whose variance is greater than some level of neighborhood variance. Specifically, for the gth feature, we can compute its variance within a neighborhood and compare it to the overall variance. The neighborhood variance is defined as,

$$S_g^{2(N)} = \frac{1}{nk_c - 1} \sum_{i=1}^{n} \sum_{j=1}^{k_c} \left(e_{ig} - e_{N(i,j)g}\right)^2$$

- k_c is the number of nearest neighbors needed for each node for the graph to be connected.
- Each e_{ig} is representing the value of feature g in cell i.
- $e_{N(i,j)g}$ is representing the feature value of the jth nearest neighbor in cell i.

Selecting Genes Most Likely to Be Involved in a Trajectory

Given computed overall variance, $\hat{\sigma}_g$ and neighborhood variance $S_g^{2(N)}$, we seek genes varying more globally than within a neighborhood so that $\hat{\sigma}_g > S_g^{2(N)}$.

Local Linear Embedding ($d = 2$)

Step 1: Find the weights (w_{ij}s) that can best reconstruct the original data (e.g. the Es cell $\times$ feature) in terms of k nearest neighbors as,

$$W = \text{argmin}_W \sum_{i=1}^{n} \left| E_i - \sum_{j=1}^{k} w_{ij} E_j \right|_2^2$$

Step 2: Find optimal d-dimensional embedding, so in this case, L

$$L = \text{argmin}_L \sum_{i=1}^{n} \left| L_i - \sum_{j=1}^{k} w_{ij} L_j \right|_2^2$$

k-NN graph and shortest path

- Compute k-nearest neighbor graph between cells in terms of the LLE-determined coordinates.
- Specify a starting point (like a stem cell), and use a shortest path algorithm like Dijkstra to find the shortest path to some cell of interest.

Detecting Branches with Geodesic Entropy Measure

- Let $t_i = \{s = v_1, \ldots, v_k, \ldots, v_l = i\}$ be the shortest path from the starting point s to some cell, i.
- Denote the kth node on the shortest path from s to i by $t_i(k)$.
- Define f_{jk} as the number of paths passing through point j at distance k, $f_{jk} = \sum_i^n I[t_i(k) = j]$
- Then compute the fraction of all paths in S that pass though node j at distance k, $p_{jk} = \frac{f_{jk}}{\sum_{i=1}^n f_{ik}}$
- $H_k = -\sum_{i=1}^n p_{ik} \log_2 p_{ik} \rightarrow$ look at high entropy

SLICER Applied to Synthetic Data

Studying geodesic entropy over k. Higher entropy in terms of steps corresponds to the 'bubbles' in the data.

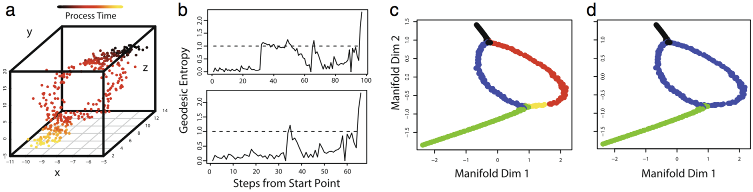


Figure: from Welch *et al.* Genome Biology. 2016.

Neural Stem-Cell Differentiation Data

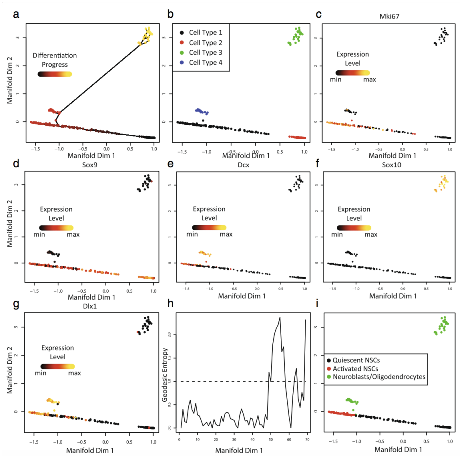


Figure: from Welch *et al.* Genome Biology. 2016.